Mohamed Sarhan

Software Architect & Red Team Specialist

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Professional Summary

Advanced **Software Engineer** and **Red Team Operator** with over 9 years of coding experience. Unlike typical developers, I specialize in engineering **custom security tools** and offensive frameworks from scratch rather than relying solely on pre-built scanners. Expert in **Embedded Systems** integration (Raspberry Pi/Robotics) and building secure, scalable backend architectures. Proven history of working **on-site** with corporate teams to deliver production-grade software.

Experience

2023–Present **Software Engineer (Embedded Systems & R&D)**, *Horus University (CIC Center)*, Damietta

- **Role:** Lead Software Engineer under the supervision of **Prof. Omar Hussein**.
- **Achievement:** Architected the control logic for an autonomous "Smart Car" prototype using **Raspberry Pi** and Python.
- Developed low-level drivers to interface software with hardware sensors, ensuring real-time response.
- Led a technical team in the R&D lab to integrate computer vision algorithms for obstacle detection.

2022–2023 **Full Stack Developer (On-site)**, *Athr*, Headquarters

- Worked on-site to architect the backend infrastructure for high-traffic web applications.
- Collaborated directly with the product team to transform business requirements into technical blueprints.
- Conducted internal security audits (Red Teaming) to harden the application against OWASP Top 10 vulnerabilities.

2021–2022 **Full Stack Developer (On-site)**, *iBrand*, Headquarters

- Developed and maintained the core frontend architecture using **React.js**, ensuring high performance.
- Optimized legacy codebases, reducing server load by 20% and improving page load speeds.
- Delivered secure, user-centric features in an Agile environment.

2020–Present **Red Team Operator & Security Researcher**, *Freelance / Independent*, Global

- **Custom Tooling:** Developed proprietary Python-based offensive tools for penetration testing, bypassing standard firewall detections.
- **Environment:** Expert in utilizing modified Windows environments for offensive operations while maintaining deep proficiency in Linux administration.
- Conducted black-box penetration testing, reporting critical vulnerabilities (SQLi, XSS, RCE) with remediation strategies.

Key Engineering Projects

GitHub **Custom Machine Learning Core**, *Python*, *Math*
https://github.com/Mohamed2007Sarhan/machine-learning

- Engineered a custom **Radial Basis Function** library from scratch without external ML frameworks.
- Solved performance limitations found in standard libraries by optimizing mathematical algorithms manually.

GitHub **Codeflow Website Platform**, *React.js*, *Node.js*
https://github.com/Mohamed2007Sarhan/codeflow-website

- Designed and built a full-stack platform for coding education/workflow.
- Implemented secure authentication and real-time data handling.

GitHub **iBrand Frontend Architecture**, *React.js*, *Redux*
https://github.com/Mohamed2007Sarhan/ibrand_front_react
- The production-grade frontend repository for iBrand.
- Showcases clean component architecture, state management.

GitHub Advanced Replace-by-fee (RBF), *React.js*, *Redux*
https://github.com/Mohamed2007Sarhan/RBF
- RBF Offensive Quality Hacking Tool Repository.
- An easy, fast, and advanced user interface for RBF work.

Technical Skills

Red Teaming	Custom Exploit Development, Offensive Security, Windows/Linux Hardening
Architecture	System Design, Embedded Systems (Raspberry Pi), CI/CD, Hardware Interfacing
Languages	Python (Expert), C++, C#, Java, JavaScript/TypeScript, SQL
Frameworks	React.js, Next.js, Node.js, FastAPI, Docker
Tools	Git, Burp Suite, Metasploit, Nmap (and Custom Scripts)

Education

2024–2029 **Bachelor of Dentistry**, *King Salman International University*, Sinai
Demonstrating exceptional discipline and multi-disciplinary academic excellence.

Certifications

Security	Certified Ethical Hacker Training / Attack Strategies
Programming	Python Advanced Course
Development	Java Programming & Web Development (WordPress)